

6.1 Computer Security and Cyber Security

Computer security means dealing with the security of a computer's software and hardware. One of the most important aspects of computer security is keeping stand-alone machines updated and properly patched.

Cyber security, on the other hand, involves the security of individual computers plus the network they belong to and the intercommunication between all the computers in an organization. It can be easy to remember the difference by knowing that the meaning of the word 'cyber' is a 'culture of computers.' It encompasses everything that is connected.

Common Security Threats

There is always an ever-evolving list of security threats and dangers. Some of the most common cyber threats are:

Viruses are malicious programs that get loaded onto victims' computers without their knowledge or permission. Typically, they replicate and infect files and programs installed on the computer.

Worms are software programs that will copy themselves from one computer to another, automatically. They use up disk space and can quickly replicate with larger volumes and speed.

Keylogger or a keystroke logger tracks the real-time activity of someone using their computer. It records every keystroke made on a keyboard and is a powerful threat as it can steal individuals' usernames, passwords and login credentials.

Botnet are when a group of computers connected online have all been compromised by a hacker using a virus. The victims' computers are typically then used for malicious activities and bigger attacks such as denial-of-service attacks.

Phishing is when criminal disguises themselves as a business or individual to steal sensitive data or personal information, typically via fraudulent emails and malicious links. Phishing is so dangerous because it is easy to execute and hackers are getting more sophisticated at making their emails look legitimate.

6.2 Information Systems

Information systems incorporate computers to help manage data and achieve business objectives, they do not necessarily have to include computers. There are different types of information systems that can serve a variety of purposes depending on an organization's needs. Examples include:

- **Data warehouses:** Data warehouses are data management systems that support analytics and other business intelligence activities. They consolidate and analyze data from a large variety of sources. Data warehouses can provide insight into businesses to help improve decision-making.
- **Enterprise systems:** Enterprise systems, also known as enterprise resource planning (ERP) systems, are integrated systems that combine all the hardware and software a business uses for different functions in its operations. These organization-wide systems help information flow between departments and allow processes from different parts of the business to be integrated across a company.
- **Expert systems:** Expert systems use artificial intelligence to mimic human decision-making. The software uses human knowledge to solve problems that would typically require a person's expertise. Expert systems can be applied in areas such as medical diagnoses, accounting and coding.
- **Geographic information systems:** Geographic information systems (GIS) are tools that gather, organize, map and analyze data with a spatial component. GIS can improve analysis and decision-making by allowing users to visualize data on a map. Global information systems are a type of GIS that synthesize worldwide data.
- **Office automation systems:** Office automation systems combine communication technology, people and computers to help perform office activities, such as preparing written communication, printing, scheduling or creating reports.

6.2.1 Components of Information Systems

Every information system includes several key components: hardware, software, telecommunications, people and data.

- **Hardware** refers to the physical pieces of the information system
- **Software** is the programming that controls the information system
- **Telecommunication** transmits information through the system
- **Humans** manage and interact with the information system
- **Data** is information stored within and processed by the system.

Hardware

The hardware component of an information system comprises the physical elements of the system. People can touch and feel pieces of hardware. These mechanisms, equipment and wiring allow systems like computers, smartphones and tablets to function.

Input and output devices are essential pieces of technology that allow humans to interact with computers and other information systems. Keyboards, mice, microphones and scanners are all examples of input devices.

Output devices might include printers, monitors, speakers and sound and video cards. Pieces of hardware including microprocessors, hard drives, electric power supply units, and removable storage also allow computers to store and process data.

Software

Software are the intangible programs that manage information system functions, including input, output, processing and storage. System software – such as the MacOS or Microsoft Windows operating systems – provides a base for application software to run. Application software operates programs for particular uses in information systems. For instance, word processing applications are used to create and edit text documents. Graphical user interface (GUI) software is among the most common application software. It presents the information stored in computers and allows users to interact with computers through digital graphics – such as icons, buttons and scroll bars – rather than through text-based commands.

Software can be either open source or closed source. Open-source software coding is publicly available for users and programmers to manipulate, whereas closed source software is proprietary.

Telecommunications

Telecommunications systems connect computer networks and allow information to be transmitted through them. Telecommunications networks also allow computers and storage services to access information from the cloud.

There are a number of methods telecommunications networks use to convey information. Coaxial cables and fiber optic cables are used by telephone, internet and cable providers to transmit data, video and audio messages.

Local-area networks (LANs) connect computers to create computer networks in a designated space, like a school or home. Wide-area networks (WANs) are collections of LANs that facilitate data-sharing across large areas. A virtual private network (VPN) allows a user to protect their online privacy by encrypting data on public networks. Microwaves and radio waves can also be used to transmit information in telecommunications networks.

Data

Data are intangible, raw facts that are stored, transmitted, analysed and processed by other components of information systems. Data are often stored as numerical facts, and they represent quantitative or qualitative information. Data can be stored in a database or data warehouse, in a form that best suits the organization using it.

Databases contain collections of data that can be queried or retrieved for specific purposes. Databases allow users to perform fundamental operations, such as storage and retrieval. Data warehouses, on the other hand, store data from multiple sources for analytical purposes. They allow users to assess an organization or its operations.

Human Resources

Human resources are a crucial part of information systems. The human component of information systems encompasses the qualified people who influence and manipulate the data in information systems. Humans involved in information systems may

include business analysts, information security analysts or system analysts. Business analysts work to elevate an organization's operations and processes. They often focus on improving efficiency and productivity or streamlining distribution. Information security analysts work to prevent data breaches and cybersecurity attacks.

6.3 Importance of Computer Security

Computer Security is, basically the preservation of computing systems and the information that they save and / or retrieve. Computer Security is important as it enables people to perform their work and study. It helps in the execution of essential business processes, and safeguards confidential and sensitive information.

Attack Vector

An attack vector is a path or means by which an attacker or hacker can gain access to a computer or network server in order to deliver a payload or malicious outcome. Attack vectors enable hackers to exploit system vulnerabilities, including the human element. Common cyber attack vectors include viruses and malware (intrusive software), email attachments, webpages, pop-up windows, instant messages (IMs), chatrooms and deception. Deception is when a human operator is fooled into removing or weakening system defences. Firewalls and antivirus software can block attack vectors. But no protection method is totally attack-proof.

6.4 Losses in case of Security Attacks

Productivity Loss

Losses that result from an organization's inability to deliver its products or services.

Response Loss

Losses that are associated with managing the event itself. This form of loss will be the most common across your analyses.

Replacement Loss

The costs associated with the replacement of a capital asset or a person.

Fines and Judgments

Penalties levied against an organization through civil, criminal or contractual actions, usually the result of a Confidentiality related scenario.

Reputation Damage

Losses associated with an external actor's perception of your organization whereby its main value proposition is diminished.

6.5 Difference between hacker and a Cracker

There are a lot of dissimilarities between hackers and crackers, and one should not use them interchangeably. The main difference between hackers and crackers can be observed through the following points:

- Hackers are people who use their knowledge for a good purpose and do not damage the data, whereas a cracker is someone who breaks into the system with a malicious purpose and damages data intentionally.
- Hackers possess advanced knowledge of computer systems and programming languages, while crackers might not necessarily be so skilled and well-versed with computing knowledge.
- The hackers work for an organization to improvise their network and solve any issues. Crackers are someone from whom the hacker protects the organization. Crackers work just because a system might be challenging or to get illegal gains.
- Hacking is ethical, while cracking is illegal and unethical.
- Hackers have ethical certificates, while the Crackers do not possess any certificates.
- Hackers continuously work towards making new tools rather than using the existing ones. The crackers, on the other hand, have inadequate computing knowledge to make new tools and use tools already used by other crackers.

Do you know

- In common language, the word *hacker* is mostly used instead of *cracker*. In this book, the word *hacker* is used throughout instead of *cracker*. While, Section 6.5 describes their differences.

Risk vs. threat vs. vulnerability

Risk is any possibility for loss, damage or destruction of assets or data caused by a cyber threat. **Threat** is a process that magnifies the likelihood of a negative event, such as the exploit of a vulnerability. A **vulnerability** is a weakness in your infrastructure, networks or applications that potentially exposes you to threats. When a threat targets a vulnerability that exists in your IT infrastructure, network or applications, it can result in risk to your assets, data or business.

Risk: An organization's risk profile fluctuates depending on internal and external environmental factors. It incorporates not just the potential or probability of a negative event, but the impact that event may have on your infrastructure. And though risk can never be 100% eliminated—cybersecurity is a persistently moving target, after all—it can be managed to a level that satisfies your organization's tolerance for risk.

Threat: Today's cybersecurity landscape roils with an endless stream of potential threats—from malware that plants dangerous executables in your software and ransomware that locks up your systems to specially targeted hacker attacks. All of these threats look for a way in, a vulnerability in your environment that they can exploit. Some threats, however, hold more potential for exploitation than others.

Vulnerability: Vulnerabilities are weak spots within your environment and your assets—weaknesses that open you up to potential threats and increased risk. And unfortunately, an organization can have thousands, often millions of vulnerabilities.

6.6 Features of a Secure System

Cybersecurity is the responsibility of every employee who has or accesses a computer that belongs to the company. Standardized rules are essential so that everyone knows what is expected of them. Here are some critical features of a sound cybersecurity policy:

Confidentiality

Confidentiality means protecting the information from being exposed to an unauthorized party due to a data breach or insider threat. Nobody wants to deal with the fallout of a data breach, which is why you should take major steps to implement document security,

establish security controls for sensitive files, and establish clear information security policies regarding devices. Confidentiality covers a spectrum of access controls and measures that protect your information from getting misused by any unauthorized access. The ideal way to keep your data confidential and prevent a data breach is to implement safeguards.

A failure to maintain confidentiality means that someone who shouldn't have access has managed to get access to private information. Through intentional behaviour or by accident. A failure in confidentiality can cause some serious devastation.

Some information security basics to keep your data confidential are:

1. Encryption
2. Password
3. Two-factor authentication
4. Biometric verification

Integrity

Integrity refers to the accuracy and completeness of data. Security controls focused on integrity are designed to prevent data from being modified or misused by an unauthorized party. Integrity involves maintaining the consistency and trustworthiness of data over its entire life cycle. Data must not be changed in transit, and precautionary steps must be taken to ensure that data cannot be altered by unauthorized people. For example, in a data breach that compromises integrity, a hacker may seize data and modify it before sending it on to the intended recipient. Some security controls designed to maintain the integrity of information include:

1. Encryption
2. User access controls
3. Version control
4. Backup and recovery procedures
5. Error detection software

Availability

Data availability means that information is accessible to authorized users. It provides an assurance that your system and data can be accessed by authenticated users whenever they're needed. Similar to confidentiality and integrity, availability also holds great value. Availability is typically associated with reliability and system uptime, which can be impacted by non-malicious issues like hardware failures, unscheduled software downtime, and human error, or malicious issues like cyberattacks and insider threats. If the network goes down unexpectedly, users will not be able to access essential data and applications.

Making regular off-site backups can limit the damage caused to hard drives by natural disasters or server failure. Information only has value if the right people can access it at the right time. Information security measures for mitigating threats to data availability include:

1. Off-site backups
2. Disaster recovery
3. Redundancy
4. Failover
5. Proper monitoring
6. Environmental controls
7. Virtualization
8. Server clustering

Authentication

Authentication is the process of identifying users that request access to a system, network, or device. Access control determines user identity according to credentials like username and password. Other authentication technologies like biometrics and authentication apps are also used to authenticate user identity.

Non-repudiation

Non-repudiation is the assurance that someone cannot deny the validity of something. Non-repudiation is a legal concept that is widely used in information security and refers to a service, which provides proof of the origin of data and the integrity of the data.

Digital signatures (combined with other measures) can offer non-repudiation when it comes to online transactions, where it is crucial to ensure that a party to a contract or a communication can't deny the authenticity of their signature on a document or sending the communication in the first place. In this context, non-repudiation refers to the ability to ensure that a party to a contract or a communication must accept the authenticity of their signature on a document or the sending of a message.

Accountability

Accountability is an essential part of an **information security plan**. The phrase means that every individual who works with an information system should have specific responsibilities for information assurance. One example would be a policy statement that all employees must avoid installing outside software on a company-owned information infrastructure.

6.7 Computer Protection

Computer protection or security is the process of protecting your computer against unauthorized intrusions. This unauthorized intrusion includes installation of malicious software on your computer without your consent and hackers taking over your computer's control so that it can be used as a botnet.

Computer Protection Basics

Once your PC security has been compromised, hackers can also take over your identity and use it to their advantage. Like, for example, hackers can pose as 'you' to carry out security attacks, can hide their actual location in case of high profile attacks and use yours instead, and can also encrypt your computer's data or crash it altogether. Some basic steps to protect a computer are:

1. **Install Antivirus Software:** Not just any antivirus software but one which can offer real-time protection. Because real-time protection is crucial if your PC is to be protected against today's security threats. Each day a new security threat is discovered and your antivirus software should keep up with them.
2. **Implement Other Security Tools:** Unfortunately, antivirus alone is not sufficient to avoid security threats. The whole package is required – anti-malware, firewall, and others – to combat today's sophisticated malware. These bunch of security tools usually are packaged and sold under the name Internet Security suite. PC users have to implement these if they need enhanced computer protection.
3. **Keep Your Computer Updated:** Ensure every software within your computer is properly patched up and up-to-date. This includes the operating system and the other third-party software which might be a part of it. Because most of the security threats exploit your system's security vulnerabilities and keeping your system up-to-date can eliminate such threats.
4. **Use Complex Passwords:** Another aspect which is very important but quite often ignored are strong and unguessable passwords. Each password you possess (not just your system password) to various accounts you might be having pose a serious threat to your PC and your digital identity.
5. **Follow Security Best Practices:** Finally, follow the security best practices like not clicking on suspicious mails, not downloading suspicious attachments, not visiting dangerous websites etc., to keep your computer and digital identity safe.

6.7.1 Antimalware vs Firewall

Firewall and antivirus are mechanisms which provide security to systems. Firewalls help control network traffic in the system by acting as barriers for incoming traffic. Whereas, antiviruses protect systems against internal attacks by perceiving or spotting malicious files and viruses.

Antivirus Software

Antivirus software is a cyber security mechanism. Its primary function is to scan, spot, and inhibit any distrustful files and software from getting into the system. Additionally, many traditional antivirus software incorporates detection of susceptibility, additional scanning of regular antivirus, and control of internet traffic.

Selection of Appropriate Antimalware?

Whether your computer or device runs Android, Mac or Windows, cybercriminals have sophisticated attacks to target machines across operating systems (OS). Choosing an antivirus software that's designed for your OS and tailored to your computing behaviours is important. Not all antivirus software is created equal, so there are following factors to consider:

1. Does your antivirus software protect you from major threats?
2. Does your program offer real-time prevention and rapid infection removal?
3. Does your software run accurately and efficiently?
4. Is your antivirus software easy to use?

6.7.2 Installing an antivirus program on a computer

To install an antivirus program on your computer, follow the steps below.

1. If you purchased the antivirus program from a retail store, insert the CD or DVD into the computer's disc drive. The installation process should start automatically, with a window opening to help guide you through the install process.
2. If you downloaded the antivirus program on the Internet, find the downloaded file on your computer. If the downloaded file is a zip file, unzip the file to extract and access the installation files. Look for a file named **setup.exe**, **install.exe**, or something similar, then double-click that file. The installation process should start, with a window opening to help guide you through the install process.
3. In the installation process window, follow the steps provided to install the antivirus program. The install process provides recommended options so the

antivirus program will function properly, which in most cases can be accepted as is.

4. When the install process is complete, close the install window.

The antivirus program is now installed and ready to use. It is recommended to restart your computer so that any modified settings in the operating system can take effect correctly.

Updating the antivirus Program

Without the latest definitions, the antivirus program will not know about the most recently created viruses and spyware, making your computer vulnerable to an infection. After installing the antivirus program, we highly recommend you update it with the latest virus and spyware definitions. The updates allow the antivirus program to protect your computer from all viruses and spyware. In many cases, the antivirus program automatically checks for and installs the latest updates. If prompted to do so, select Yes to update the antivirus program. If it does not prompt you to update immediately.

Performing a Scan

Correctly setting up and running an antivirus scan on your computer is one of the best starting defences for keeping your system free of malicious software. A strong defence begins with selecting an antivirus solution for your computer and understanding how to get the most out of it. Learning to properly use a virus scan will keep you much safer: without regular full scans, your system may be hiding some unpleasant surprises.

Virus scans search through your system to locate and remove any malicious threats on your device. You'll find most antivirus software guards against malware.

Follow these steps to run your first scan:

1. **Verify that all security updates are downloaded.** Updates will include all heuristic virus definitions and other data needed to scan properly.
2. **Execute a full virus scan.** This will scan hard drives, removable media, system memory, email and the like for viruses.
3. **Clean your backups first** by executing a virus scan before restoring any data.

4. **Read any result reports for further action.** Malware will usually be quarantined to await action from you. You'll have to decide what to do with the potential threat. Most antivirus apps give you a recommended action to simplify the decision for you.
5. **Follow prompts to remove or keep potential threats.** You'll want to remove most quarantined files usually. On occasion, you may find a clean file that has been miscategorized and given you a false positive. High-quality antivirus software should rarely label clean files as threats, so be sure you do your research on detection rates.

Teacher Notes:

- Perform an antivirus scan in class and discuss the results with students.

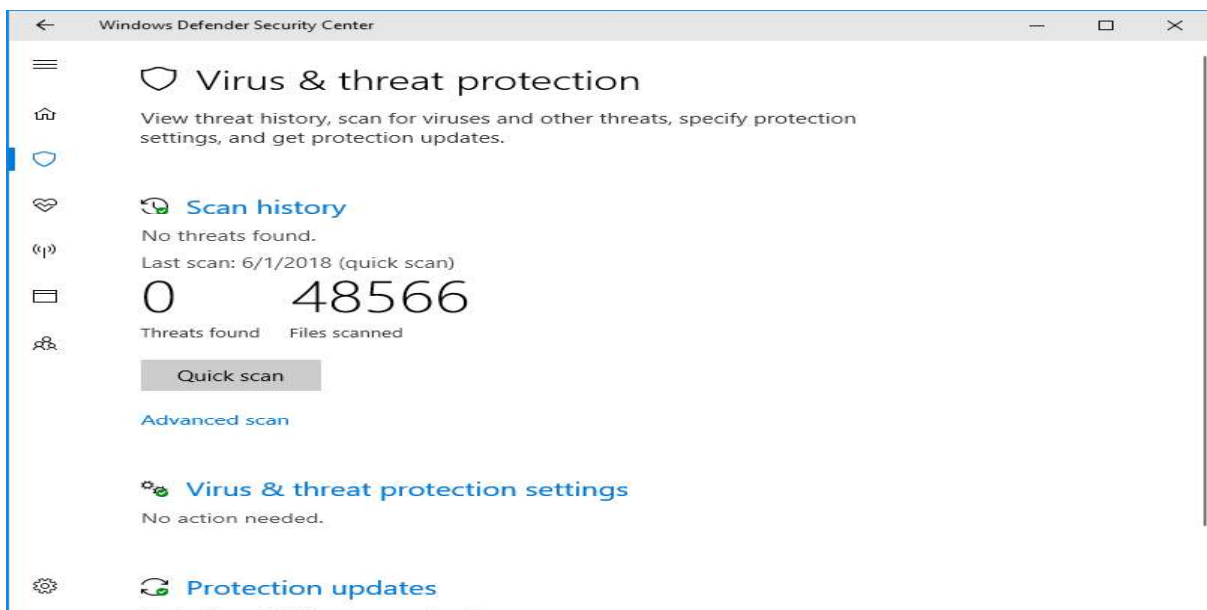


Fig. 6.1 Performing a Scan

6.8 Firewalls

A firewall is a network security device that monitors incoming and outgoing network traffic and decides whether to allow or block specific traffic based on a defined set of security rules. It establishes a barrier between secured and controlled internal networks that can be trusted and untrusted outside networks, such as the Internet. A firewall can be hardware, software, or both.

Top 10 Firewall Software

- FortiGate
- Check Point Next Generation Firewalls (NGFWs)
- Sophos XG Firewall
- Huawei Firewall
- Avast
- WatchGuard Network Security
- Palo Alto
- SonicWall
- GlassWire Firewall
- Cisco

Configuring a Firewall

Step 1: Secure your firewall

If an attacker is able to gain administrative access to your firewall it is “game over” for your network security. Therefore, securing your firewall is the first and most important step of this process. Never put a firewall into production that is not properly secured by at least the following configuration actions:

Update your firewall to the latest firmware.

- Delete, disable, or rename any default user accounts and change all default passwords. Make sure to use only complex and secure passwords.
- If multiple administrators will manage the firewall, create additional administrator accounts with limited privileges based on responsibilities. Never use shared user accounts.
- Disable simple network management protocol (SNMP) or configure it to use a secure community string.

Step 2: Architect your firewall zones and IP addresses

In order to protect the valuable assets on your network, you should first identify what the assets (for example, payment card data or patient data) are. Then plan out your network structure so that these assets can be grouped together and placed into networks (or zones) based on similar sensitivity level and function. For example, all of your servers that

provide services over the internet (web servers, email servers, virtual private network (VPN) servers, etc.) should be placed into a dedicated zone that will allow limited inbound traffic from the internet (this zone is often called a demilitarized zone or DMZ). Servers that should not be accessed directly from the internet, such as database servers, must be placed in internal server zones instead. Likewise, workstations, point of sale devices, and voice over Internet protocol (VOIP) systems can usually be placed in internal network zones. Generally speaking, the more zones you create, the more secure your network. But keep in mind that managing more zones requires additional time and resources, so you need to be careful when deciding how many network zones you want to use.

Step 3: Configure access control lists

Now that you have established your network zones and assigned them to interfaces, you should determine exactly which traffic needs to be able to flow into and out of each zone. This traffic will be permitted using firewall rules called access control lists (ACLs), which are applied to each interface or sub interface on the firewall. Make your ACLs specific to the exact source and/or destination IP addresses and port numbers whenever possible. At the end of every access control list, make sure there is a “deny all” rule to filter out all unapproved traffic. Apply both inbound and outbound ACLs to each interface and sub interface on your firewall so that only approved traffic is allowed into and out of each zone.

Step 4: Configure your other firewall services and logging

If your firewall is also capable of acting as a dynamic host configuration protocol (DHCP) server, network time protocol (NTP) server, intrusion prevention system (IPS), etc., then go ahead and configure the services you wish to use. Disable all the extra services that you don’t intend to use. Configure your firewall to report to your logging server, and make sure that enough details are included.

Step 5: Test your firewall configuration

In a test environment, verify that your firewall works as intended. Don’t forget to verify that your firewall is blocking traffic that should be blocked according to your

configurations. Testing your firewall should include both vulnerability scanning and penetration testing. Once you have finished testing your firewall, your firewall should be ready for production.

Logging Services in Firewall

A firewall, at its most basic form, is created to stop connections from suspicious networks. It inspects the source address (of data packet), destination address, and the destination port of all connections, and decides if a network can be trusted. For simplicity, we can aggregate information on the source address, source port, and destination address and port. The success of any firewall typically relies on the rules used to configure it. To be most effective, a firewall ruleset must be augmented with a successful logging feature. The logging feature documents how the firewall deals with traffic types. These logs offer insights into, for example, source and destination IP addresses, protocols, and port numbers.

Uses of Firewall

- To see if new firewall rules work well or to debug them if they do not work properly.
- To discover if any malicious activity is occurring within your network. However, it doesn't offer the information you need to isolate the source of the activity.
- If you identify repeated unsuccessful tries to access your firewall from a single IP address (or from a group of IP addresses), you may wish to create a rule to stop all connections from that IP.
- Outgoing connections derived from internal servers, for example, web servers, may show that someone is using your system as a launch pad. They could be launching attacks against computers on other networks from your system.

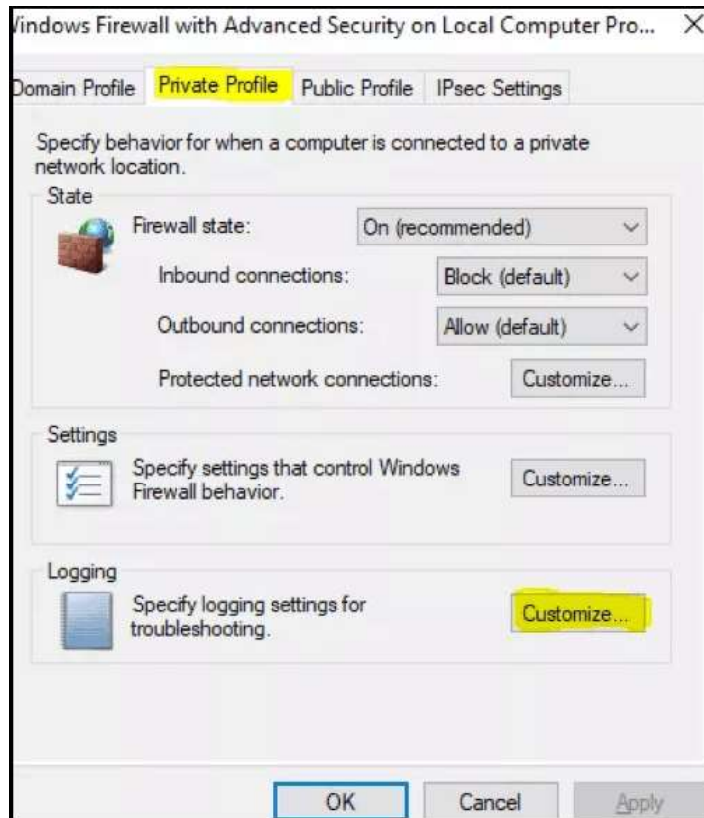


Fig. 6.2 Windows Firewall

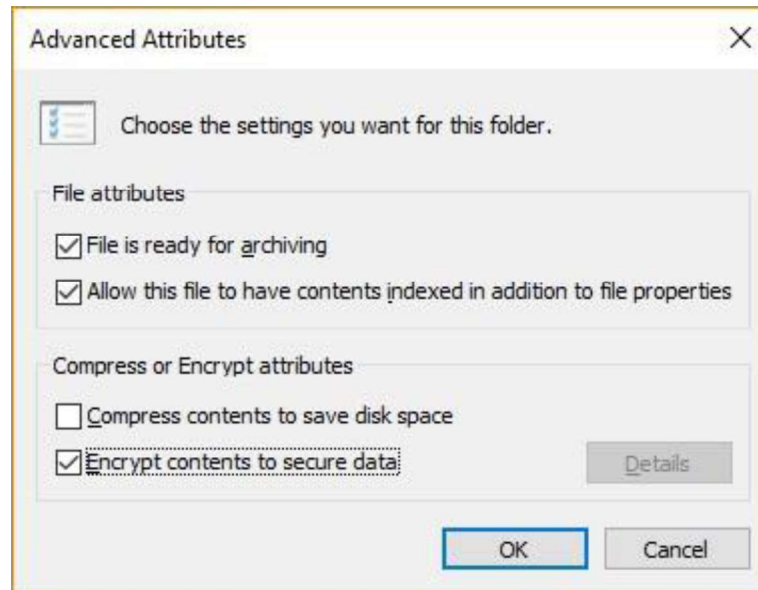
6.9 Encryption and Decryption

Encryption is the process of translating plain text data (*plaintext*) into something that appears to be random and meaningless (*ciphertext*). Decryption is the process of converting ciphertext back into plaintext.

Encrypting a file in Windows 10

File encryption helps protect your data by encrypting it. Only someone with the right encryption key (such as a password) can decrypt it. File encryption is not available in Windows 10 Home.

1. Right-click (or press and hold) a file or folder and select **Properties**.
2. Select the **Advanced** button and select the **Encrypt contents to secure data** check box.
3. Select **OK** to close the **Advanced Attributes** window, select **Apply**, and then select **OK**.



6.10 Backup and Restoration

Backup is the process of making a copy of information. The copy is typically transferred to a backup medium; i.e., either a virtual tape file, which is an allocation of disk space, or any physical backup device supported by Windows. Making backups of files on backup media is an important part of protecting data and ensuring the integrity of files. If a file is subsequently deleted by accident or if its data becomes corrupted, you can restore the most recent version of the file from backup media. Generally speaking, the system administrator at each site should establish routine backup procedures for the entire system. However, application programmers should know enough about these procedures to understand what is being backed up and with what frequency.

Creating Back up of Windows

There are several ways to back up your PC.

1. Select the **Start** button, then select **Control Panel > System and Maintenance > Backup and Restore**.
2. Do one of the following:
 - If you've never used Windows Backup before, or recently upgraded your version of Windows, select **Set up backup**, and then follow the steps in the wizard.

- If you've created a backup before, you can wait for your regularly scheduled backup to occur, or you can manually create a new backup by selecting **Back up now**.
- If you've created a backup before, but want to make a new, full backup rather than updating the old one, select **Create new, full backup**, and then follow the steps in the wizard.

Restore your files with File History

If you're missing an important file or folder that you've backed up, here's how to get it back:

1. In the search box on the taskbar, type **restore files**, and then select **Restore your files with File History**.
2. Look for the file you need, then use the arrows to see all its versions.
3. When you find the version you want, select **Restore** to save it in its original location. To save it in a different place, right-click **Restore**, select **Restore to**, and then choose a new location.

Find your backups created on previous versions of Windows

If you use *Backup and Restore* to back up files or create system image backups in previous versions of Windows, your old backup is still available in Windows 10. In the search box on the taskbar, type **control panel**. Then select **Control Panel > System and Security > Backup and Restore (Windows 7)**.

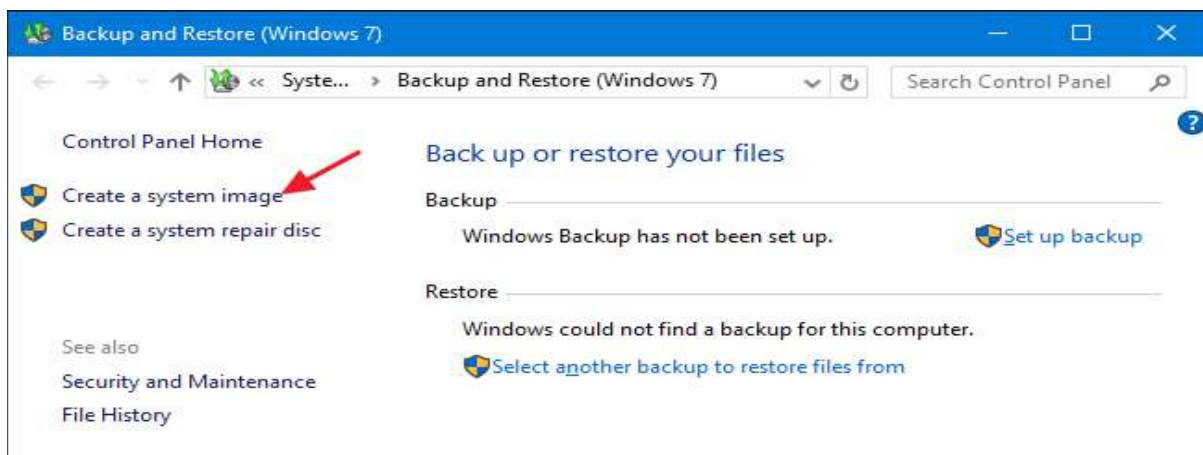


Fig. 6.3 Backup and Restore (Windows)

Key Points

- Computer security means dealing with the security of a standalone computer's software and hardware.
- Cyber security involves the security of individual computers plus the network they belong to and the intercommunication between all the computers in your organization, mobile devices, printers and more.
- Viruses are malicious programs that get loaded onto victims' computers without their knowledge or permission.
- Worms are software programs that will copy themselves from one computer to another, automatically.
- An attack vector is a path or means by which an attacker or hacker can gain access to a computer or network server in order to deliver a payload or malicious outcome.
- Hackers are people who use their knowledge for a good purpose and do not damage the data, whereas a cracker is someone who breaks into the system with a malicious purpose and damages data intentionally.
- Risk is the potential for loss, damage or destruction of assets or data caused by a cyber threat.
- Threat is a process that magnifies the likelihood of a negative event, such as the exploit of a vulnerability.
- A vulnerability is a weakness in your infrastructure, networks or applications that potentially exposes you to threats.
- Authentication is the process of identifying users that request access to a system, network, or device.
- Firewalls help control network traffic in the system by acting as barriers for incoming traffic.
- Antiviruses protect systems against internal attacks by perceiving or spotting malicious files and viruses.

Exercise

Select the most appropriate option

1. Which of the following is a component of an Information System?
a. Hardware b. Software c. Telecommunication d. All of these
2. _____ are the people who use their knowledge for good purpose.
a. Intruders b. Hackers c. Crackers d. Both a and b
3. _____ is a weakness in your infrastructure.
a. Risk b. Threat c. Vulnerability d. None of these
4. _____ controls network traffic.
a. Antivirus b. Firewall c. Wireshark d. Both a and b
5. _____ is a process of making a copy of information.
a. scanning b. restore c. backup d. Both a and b

Answer the following question.

1. Define Computer Security.
2. Differentiate between Computer Security and Cyber Security.
3. Define threats, attack vectors and vulnerability.
4. What are Information systems?
5. Describe assets of an Information system.
6. Define hacker, cracker and security expert..

7. Define confidentiality, integrity and authenticity.
8. Describe Encryption and Decryption.
9. Describe Malware.
10. Differentiate between Firewall and Anti-virus.

Practical Tasks

1. Install antimalware software on Windows.
2. Perform a malware scan on Windows.
3. Install and configure a firewall on Windows.
4. Configure firewall to create log files.
5. Perform a system restore on Windows.